
Central Valley Regional Water Quality Control Board

TECHNICAL MEMORANDUM

TO: George Low, PG
Senior Engineering Geologist

FROM: Ron Falkowski
Engineering Geologist

DATE: 6 September 2019

SIGNATURE: Original signed by Ron Falkowski

**SUBJECT: REVIEW OF NITRATE AND SETBACK CONDITIONS FOR
RICHARDSON SPRINGS COMMUNITY SERVICES DISTRICT (CSD), WDR ORDER
94-245, BUTTE COUNTY**

I have reviewed the case file for Richardson Springs CSD. The file documents the general condition of the wastewater treatment system and evaporation ponds. The Discharger has kept adequate maintenance documentation, treatment and disposal infrastructure appeared in good order. The dissolved oxygen levels are monitored at the first wastewater treatment pond.

The average daily wastewater flow (2017-2018) is less than 11,000 gallons per day. The three treatment ponds have a design capacity which exceeds storage volume required by the existing facility. The ponds can hold and evaporate the daily discharge through estimated seasonal wet weather periods until conditions are appropriate for discharge to the onsite Land Application Area (LAA).

POTENTIAL THREATS TO WATER QUALITY

The wastewater treatment system is located southwest of the main facility complex. The closest distance from the wastewater ponds to the nearest property line is greater than 5 feet. The closest potable water well is greater than 500 feet from the wastewater ponds. The closest surface water is less than the prescribed 150 feet setback from the two (2) primary treatment ponds. However, due to an inadequate soil depth underlain by Tuscan lava flows all onsite ponds are lined. Allowing setback limits to be waived as described by Requirements Section B.1, L, (v). of the General Order. Currently no groundwater monitoring network has been established. Completion of the Nitrate

Checklist in Attachment 1 of Order 2014-0153-DWQ indicates the following flow and rationale:

A1 Exceed 20,000 gpd? No.

Wastewater flow generally less than 11,000 gpd.

Conclusion: No nitrogen removal is required.

MONITORING REQUIREMENTS

To protect water quality, General Order monitoring requirements will be sufficient. In summary, Staff recommends quarterly reporting of the average daily flow rate, dissolved oxygen, odors, freeboard, berm conditions and annual inspections of the septic tank. Quarterly monitoring will be reported by the first day of the second month after the quarter ends (e.g. January-March report is due by May 1st). Annual monitoring will be included with the fourth quarter monitoring.